

Decision: Major Revision

Editor Comment: *Please read through this entire reviewer decision letter carefully and take all actions seriously in order to avoid any delay in the review process of your revised manuscript submission. You need to upload a 'Detailed Response to Reviewers' in the system.*

The comments of the 2 reviewers are listed below:

Reviewer 1:

1. General Comments and Suggestions::

- The study presents the design of a low-cost, webcam-based eye-tracking system capable of analyzing reader preferences. Beyond this primary application, the proposed system is structured to be extensible, allowing integration into various analytical frameworks and problem domains.

In the study, the MediaPipe library was employed for iris tracking. To mitigate noise arising from micro-movements, a smoothing approach based on the Kernel Density Estimation (KDE) method was implemented. The extracted eye coordinates were subsequently transmitted to other web-based application layers, and satisfactory performance outcomes were achieved.

From a writing perspective, the manuscript is well structured and demonstrates an adequate literature review. Nevertheless, several issues requiring clarification, revision, or further discussion are outlined below:

1. The system operates following a 20-second calibration procedure. After calibration, is the user required to maintain a fixed head position, or is head displacement tolerated? Specifically, how does the system perform when the user shifts laterally from the center of the screen? The robustness of the model against head movement should be clarified and, if possible, quantitatively evaluated.

2. It appears that the KDE method is applied to temporal gaze points. This should be explained more explicitly in the manuscript. Updating Figure 4 to better illustrate this process would improve clarity. Furthermore, does KDE provide a measurable performance advantage compared to simply computing the centroid (mean) of the gaze points? This comparison should be analyzed and discussed. For example, if a computationally simpler averaging algorithm achieves comparable performance while reducing processing time by even a few milliseconds, it may represent a more efficient alternative.

Additionally, while KDE appears to stabilize (fix) the eye center, this may introduce a potential limitation. For instance, could such smoothing mask indicators of attentional instability in individuals with focus-related difficulties? In other words, might this approach inadvertently suppress diagnostically relevant variability? This issue could be addressed in the discussion section.

Formal and typographical suggestions:

- * Page 9, Section 2.8, first paragraph: “Efficient” should be written in lowercase.
- * Page 10, paragraph 2: “(x_s^m, y_s)” — the duplicated “in” should be corrected. Similar minor typographical issues (e.g., “to,” “for,” etc.) should be reviewed.
- * Table 5: The abbreviation “CSS” should be defined. It does not appear to be introduced earlier in the manuscript.
- * Pages 12 and 13, Sections 2.12, 2.13, and 2.14: There may be a typographical error in the first sentence beginning with “The Dwell...”.
- * Page 14, paragraph 3: “This figure 9...” may contain a typographical error.
- * The font sizes of figure captions and in-figure text should be increased to improve readability.
- * Figure 3 should be provided as a proper graph with enhanced quality, and the embedded text should be resized to ensure legibility.
- * Section 3.8 title: “Distirbution” should be corrected to “Distribution.”

Reviewer 2:

The manuscript presents a practically motivated, end-to-end system for real-time, browser-based reading analysis using webcam-derived gaze signals and paragraph-level dwell-time analytics, complemented by optional AI-assisted support (e.g., summarization and text-to-speech). The integration of a Chrome extension with a companion service and an interactive UI is a clear strength and suggests potential real-world applicability. However, in its current form the paper does not yet meet publication standards due to critical gaps in ethics/privacy and consent, study protocol transparency, baseline/ablation evidence, metric/ground-truth clarity, and real-time performance reporting. The issues below must be addressed.

1) Ethics, informed consent, and data privacy protocol is missing / inconsistent

The study relies on webcam-based eye tracking and derives gaze-related signals from face/eye landmarks, which may involve biometric and/or personally sensitive data. However, the manuscript states that ethics approval is not required; in its current form, this justification is insufficient and high-risk for a human-subject study.

Missing: A verifiable ethics/consent/privacy and data governance description.

Required revision (actionable): Add a dedicated “Ethics & Privacy” subsection clarifying:

- *Informed consent: Was explicit consent obtained? From whom, how, and with what participant information sheet? Was withdrawal possible?*
- *Data type: Was raw webcam video recorded/stored, or only derived gaze coordinates / feature representations?*
- *Anonymization: Face images/landmarks may be re-identifiable; specify anonymization/privacy measures (e.g., on-device processing, no raw video storage, pseudonymization).*
- *Storage & access: Where are data stored, for how long, and who can access them? What security measures are used (encryption/access control)?*

- *Third-party transfer: If LLM/TTS or any external APIs are used, does any data leave the device? If yes, what payload is transmitted and how is it protected? If no, explicitly state that all processing is local and no data are transmitted to third parties.*

2) Data collection and participant protocol are not numerically auditable

The manuscript reports session-level outcomes and some aggregate counts, but it does not clearly specify the number of unique participants, inclusion/exclusion criteria, participant profile, device variability, and environmental conditions.

Missing: A reproducible study protocol at the level expected for empirical HCI/AI system papers.

Required revision (actionable): Add a “Data Collection / Study Protocol” summary table including:

- *#participants (unique individuals) and minimal demographics summary (e.g., age range)*
- *#sessions, session duration, reading material type/length, total paragraphs*
- *device/screen info (resolution, screen size), camera type (built-in/external), basic conditions (approx. distance/lighting)*
- *calibration procedure (9-point) and whether re-calibration/drift handling was performed*

3) Lack of webcam baseline and ablation studies

While the paper motivates webcam tracking as a low-cost alternative to hardware eye trackers, it does not provide a quantitative comparison against at least one webcam-based baseline method (same family of solutions). A conceptual comparison table alone is not sufficient.

Missing: “Proposed vs. baseline” evidence.

Required revision (actionable):

- *Add at least one webcam gaze-tracking baseline (from literature or a reasonable open-source method) and compare under the same protocol: accuracy/stability/latency.*

- *Provide minimal ablation:*

- o *KDE smoothing on vs. off*

- o *calibration variant (e.g., 5-point vs 9-point, or re-calibration strategy)*

- o *dwell-threshold strategy (single threshold vs. two-threshold/interval logic)*

4) Metric definitions and ground-truth validation are unclear

The manuscript reports metrics such as “focus score,” “difficulty ratio,” and “mean focus score,” but their formal definitions, computation pipeline, and validation approach are not sufficiently specified.

Missing: Clear metric definitions and reliability reporting.

Required revision (actionable):

- *Provide explicit formulae + a worked example for “focus score” and “difficulty ratio.”*
- *Report at least dispersion/uncertainty (mean±std or 95% CI) and per-session/per-participant distributions.*
- *If feasible, add a small-scale reference validation subset (task-based ground truth, comprehension questions, or a controlled validation protocol). At minimum, strengthen “Threats to Validity.”*

5) LLM/TTS components are underspecified (model/service, operating mode, and data flow)

The paper includes AI-assisted features such as LLM-based summarization and TTS-based reading, yet it does not specify the LLM/TTS provider/model, whether inference is local or cloud-based, what data are transmitted/logged, and what safeguards are in place.

Missing: Technical details and privacy/security implications of LLM/TTS usage.

Required revision (actionable):

- *Specify the LLM/TTS provider/model name(s) and whether they run locally or via cloud APIs*
- *If APIs are used: detail the payload (raw paragraph text? user identifiers?), masking/anonymization, and access controls*
- *Describe prompt/output constraints and logging/retention policy (if any)*

6) “Real-time” claim requires end-to-end latency distribution and hardware specification

The reported component-level performance values are helpful; however, for a strong “real-time” claim, the paper should report end-to-end latency (camera → processing → decision → UI update) and the test hardware used.

Missing: End-to-end latency statistics and system configuration.

Required revision (actionable):

- *Report end-to-end latency as a distribution, at minimum p50 and p90 (optionally p99)*
- *Provide the test machine specifications: CPU model, RAM, and OS (and GPU if used)*
- *Optionally add CPU utilization / frame drop rate to support real-time robustness*